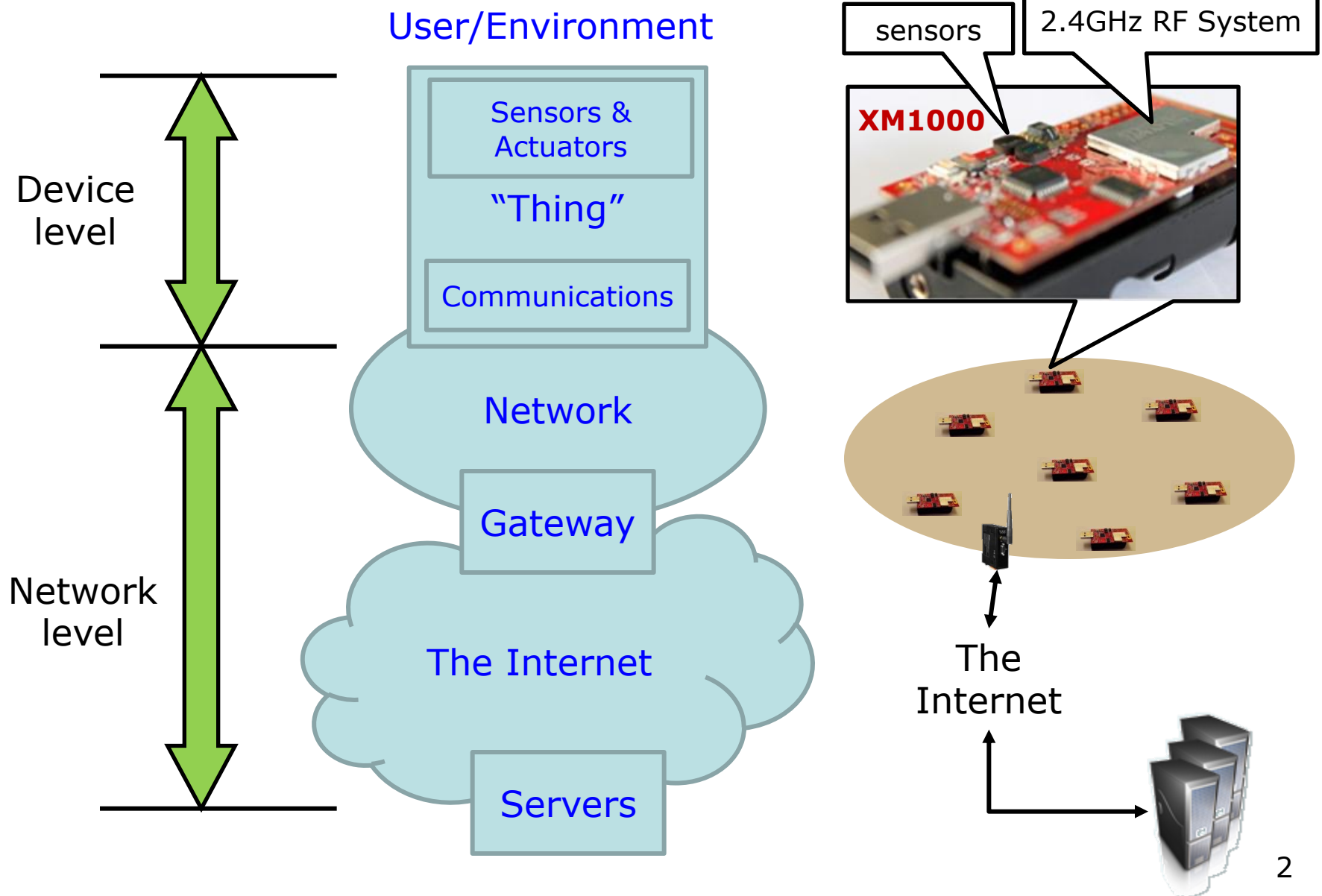


Internet of Things

Computer Science Master: IoT Curriculum

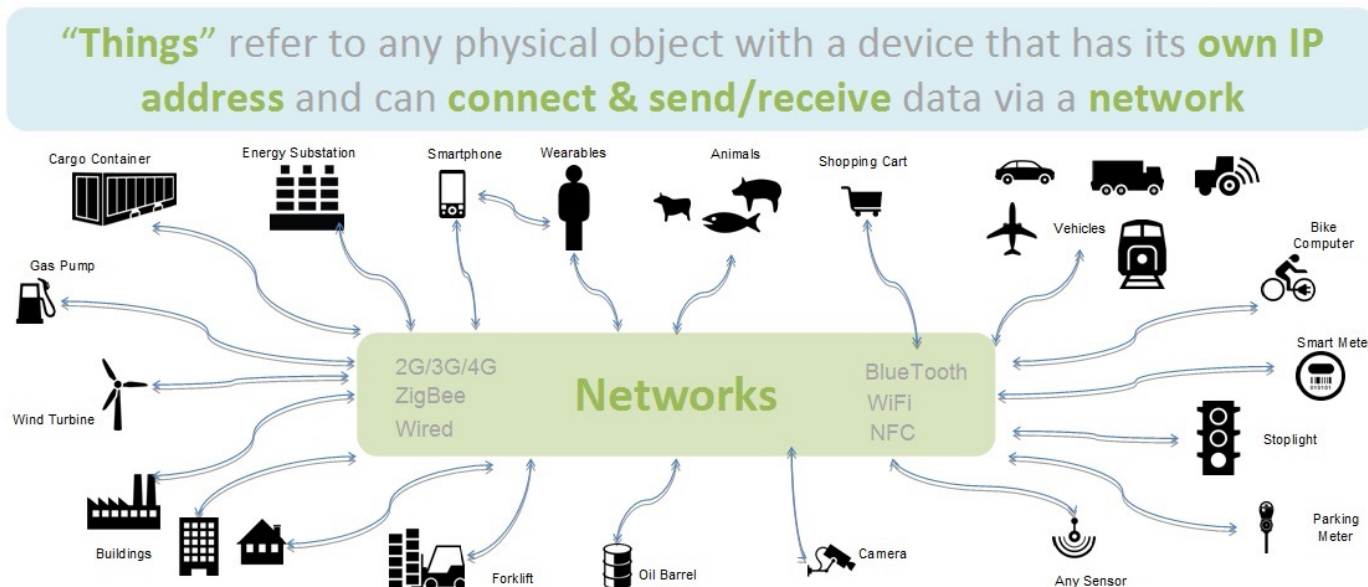
The IoT Architecture

Overview: Hardware Platform



Things

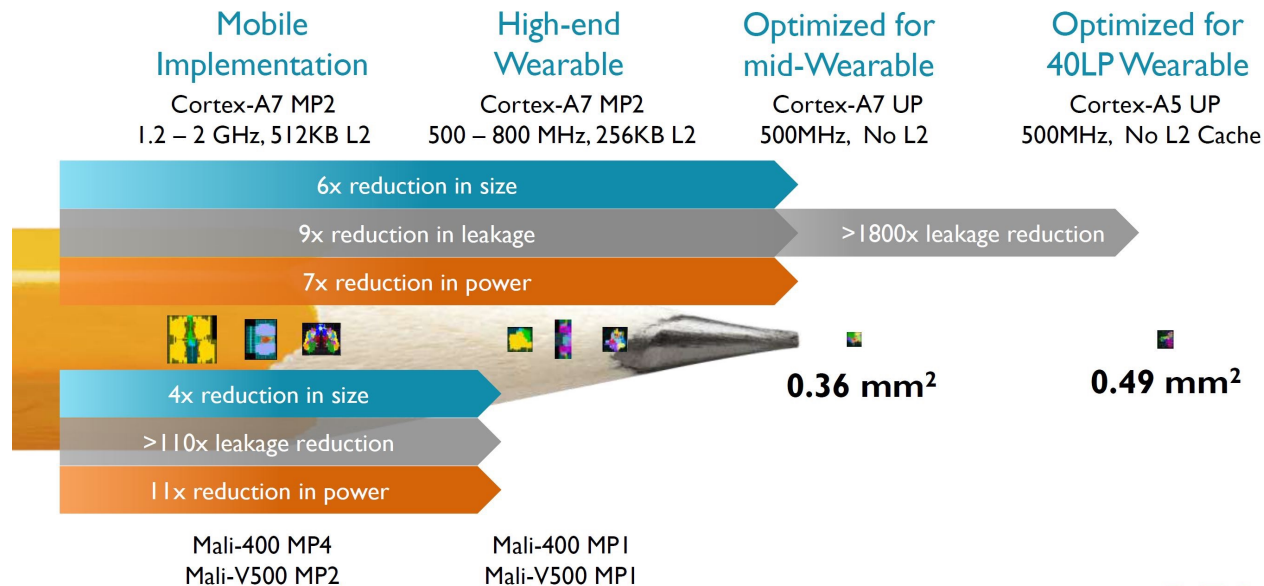
- We can turn almost every object into a “thing”.
- A “thing” still looks much like an embedded system currently.
- A “thing” generally consists of four main parts:
 - Sensors & actuators
 - Microcontroller
 - Communication unit
 - Power supply



Things

- A “thing” has the following properties:
 - It’s usually powered by battery (limited source of energy)
 - It’s small in size and low in cost. This limits computing capability.
 - It doesn’t usually perform complicated tasks.
- Power consumption is the main design issue.

Scalable Area, Performance and Battery Life



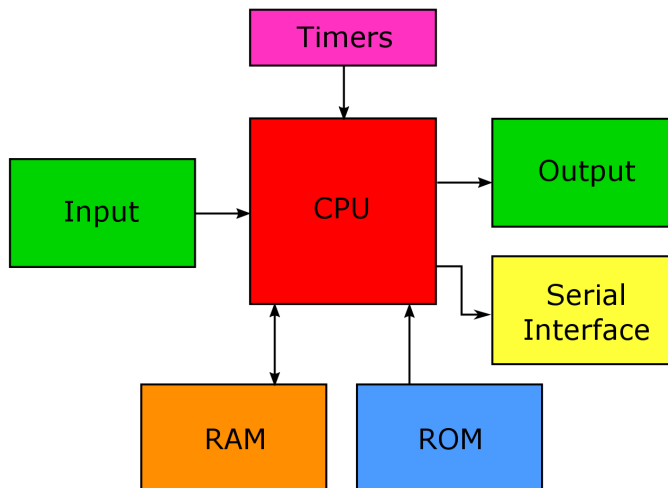
Area based on 28HPM node, 9T, unless otherwise indicated.



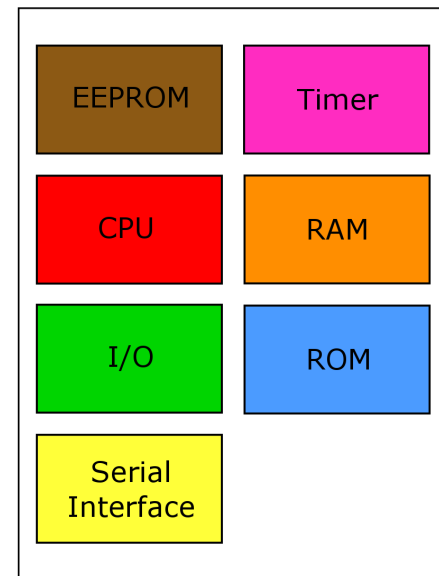
Microcontroller architecture

- A microcontroller (or MCU) is a small computer on a single integrated circuit (system on a chip or SoC) containing one or more CPUs (cores) along with memory and programmable I/O peripherals.
- It can be embedded in other machinery, such as automobiles, telephones, appliances, and peripherals for computer systems.
- Microcontrollers make it economical to digitally control even more devices and processes. Mixed signal microcontrollers are common, integrating analog components needed to control non-digital electronic systems.

Microprocessor: CPU
and several supporting chips.



Microcontroller: CPU
on a single chip.



What is a mote?

Imote2 06 with
enalab camera



- **mote** noun [C] LITERARY
something, especially a bit of dust, that is so small it is almost impossible to see
---Cambridge Advanced Learner's Dictionary
<http://dictionary.cambridge.org/define.asp?key=52014&dict=CALD>

Evolution of Sensor Hardware Platform (Berkeley), [Alec Woo 2004]

WeC 1/00



Rene 11/00



Mica 1/02



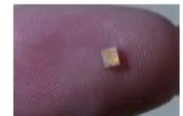
Mica2 9/02



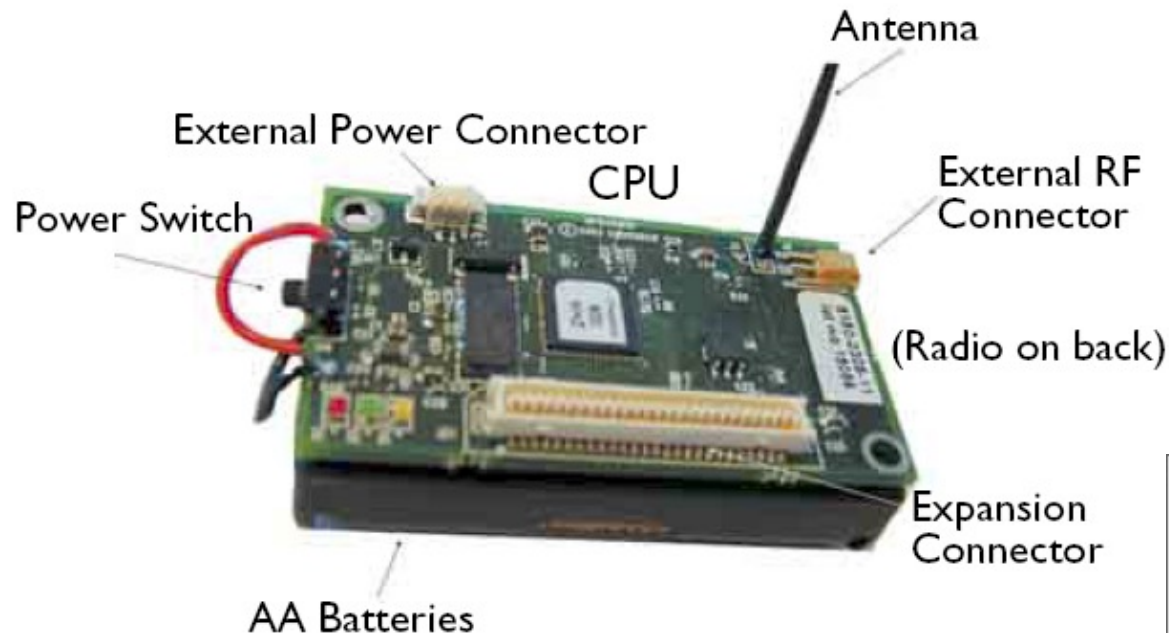
Mica2dot 9/02



SPEC 5/03



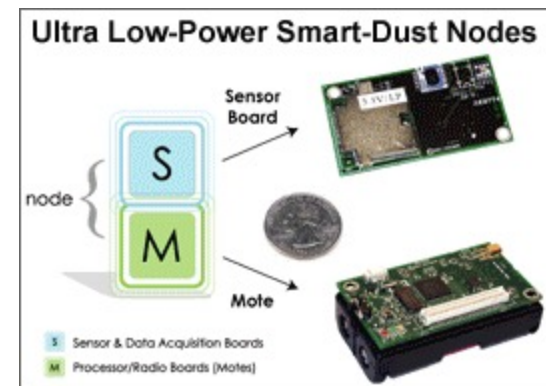
Mica2 Wireless Sensors



New MicaZ follows IEEE 802.15.4 Zigbee standard with direct sequence spread spectrum radio and 256kbps data rate

MTS310 Sensor Boards

- Acceleration,
- Magnetic,
- Light,
- Temperature,
- Acoustic,
- Sounder



[Adapted from Crossbow web site](#)

Open HW Embedded Solutions

- Raspberry PI IoT Kit, based on the super-ultra-low-cost, tiny-small-form-factor Raspberry Pi



Open HW Embedded Solutions

- **Arduino** is an open source computer hardware and software company, project, and user community

ARDUINO MKR IOT BUNDLE

Code: GKX00006



ARDUINO MKR1000 WIFI ENVIRONMENTAL MONITORING BUNDLE (AGRICULTURE & PLANTS)

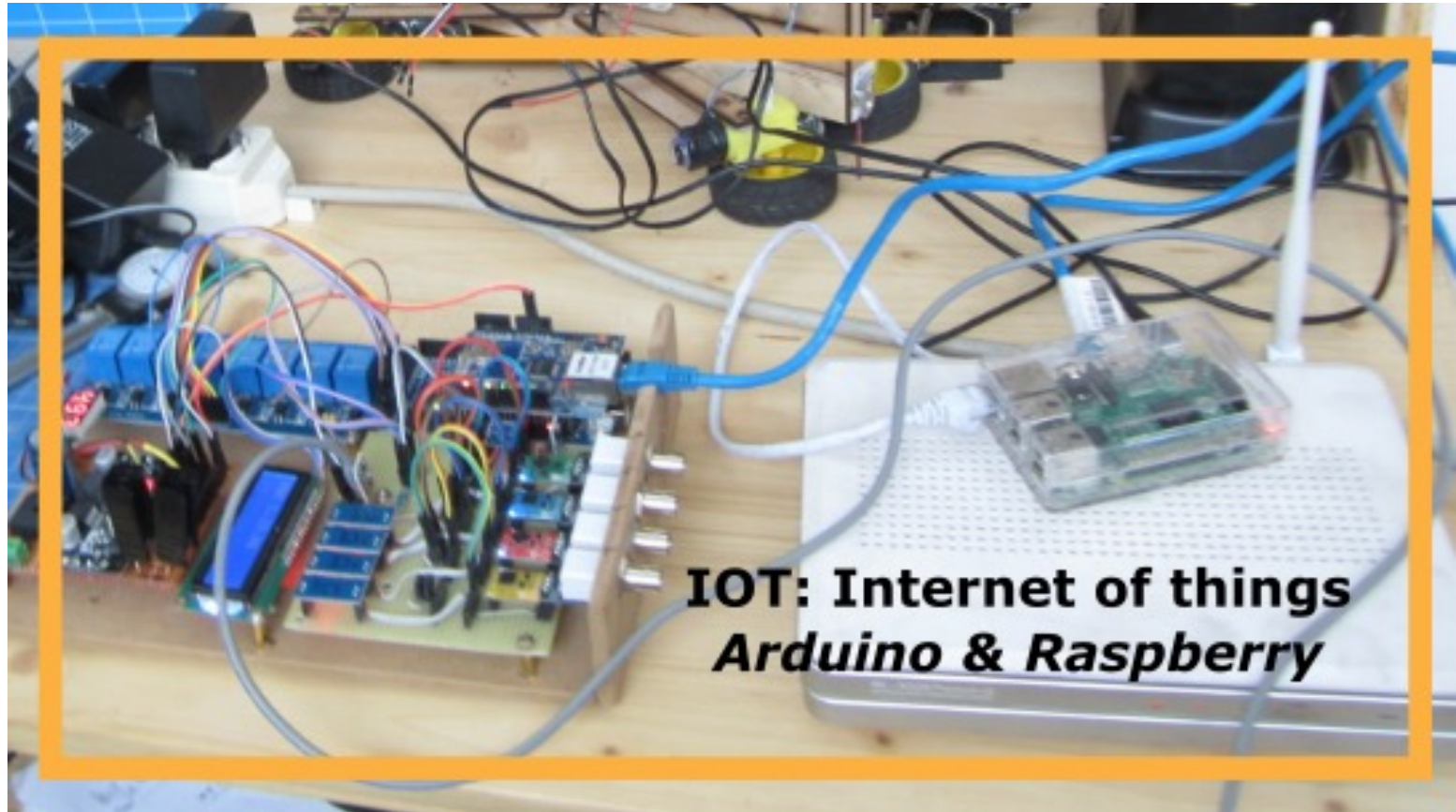
Code: VB00009

- [microcontrollers](#) and [microcontroller kits](#) based on a single-board architecture for building digital devices and interactive objects that can sense and control objects in the physical and digital world.



Open HW Embedded Solutions

- ...or combined arrangements:

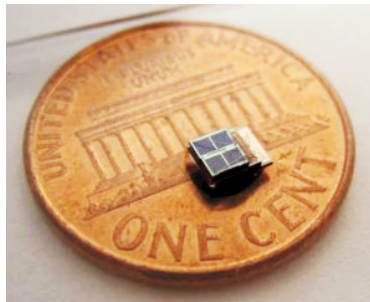


Enabling technologies

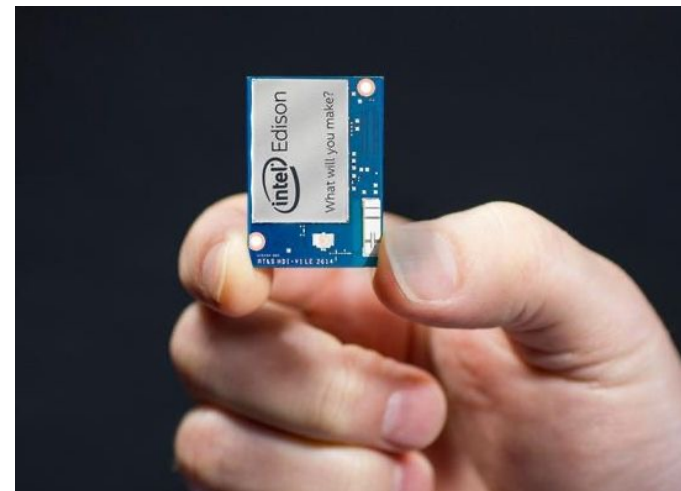
- Low-power microscopic sensors with wireless communication capability
- Miniaturization of computer hardware → Intelligence
- Micro Electro-Mechanical Structures (MEMS) → Sensing
- Low-cost CMOS-based RF Radios → Wireless Communications



Power Harvesting

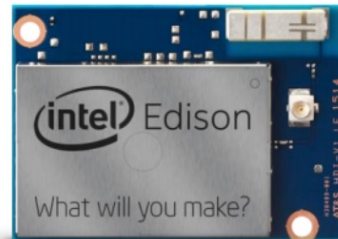


Super Capacitors



Key characteristics of IoT devices

- Often inexpensive sensors (actuators) equipped with a radio transceiver for various applications, typically low data rate $\sim 10\text{-}250$ kbps (but not always).
- Deployed in large numbers
- The sensors should coordinate to perform the desired task.
- The acquired information (periodic or event-based) is reported back to the information processing centre (or some cases in-network processing is required)
- **Solutions are often application-dependent.**



Types of IoT nodes

- **Sensor nodes**
 - Low power
 - Consist of sensing device, memory, processor and radio
 - Resource-constrained
- **Actuator nodes**
 - Like sensors but contain remote control devices
- **Sink nodes**
 - Another sensor node or a different wireless node
 - Normally more powerful/better resources
- **Gateway**
 - A more powerful node
 - Connection to core network
 - Could consist service representation, cache/storage, discovery and other functions

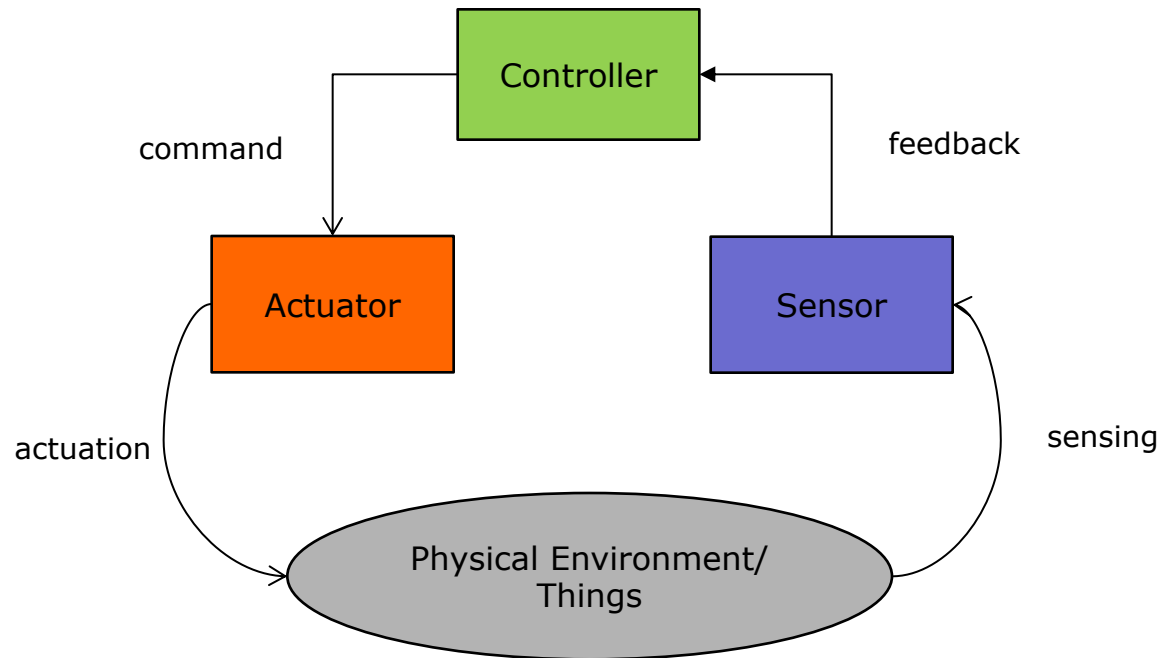
Sensors & Actuators

- Sensors:
 - mainly input components
 - sense and collect surrounding information
 - Basically three types:
 - Passive, omnidirectional (e.g. mic)
 - Passive, narrow-beam sensor (e.g. PIR)
 - Active sensors (e.g. sonar, radar, etc.)
- Actuators:
 - mainly output components
 - alter the surrounding. Some examples:
 - Adding lighting, heat, sound, etc.
 - Controlling motors to move objects
 - Displaying messages
 - and others...
- Nodes can sometimes join and leave the network due to mobility, failure etc.



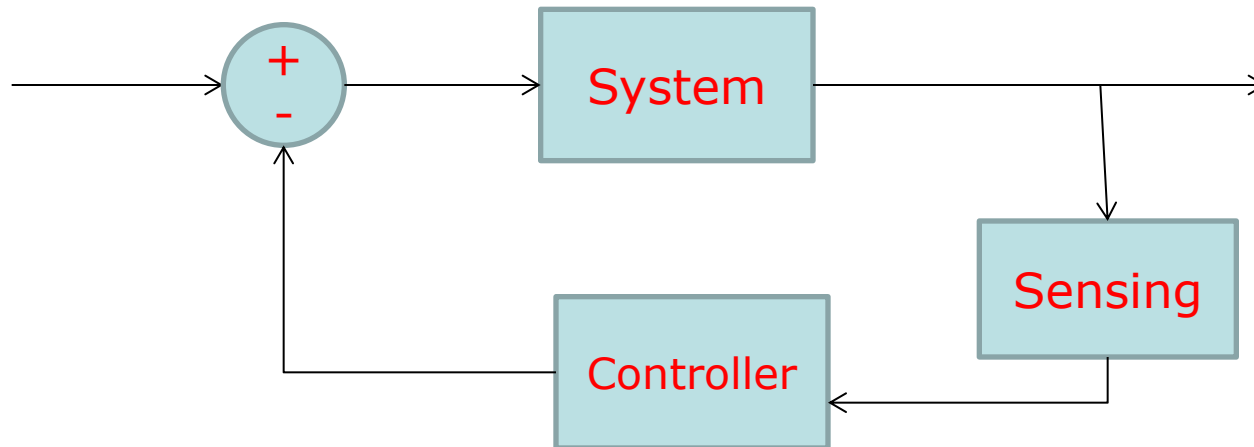
Sensors & Actuators: Control and feedback

- Control systems leverage on actuators, sensors, motors, etc.
- Sensors are used to take information from the environment
- Actuators are used to interact with the environment;
- Make a change and then sense to obtain feedback from the physical environment.



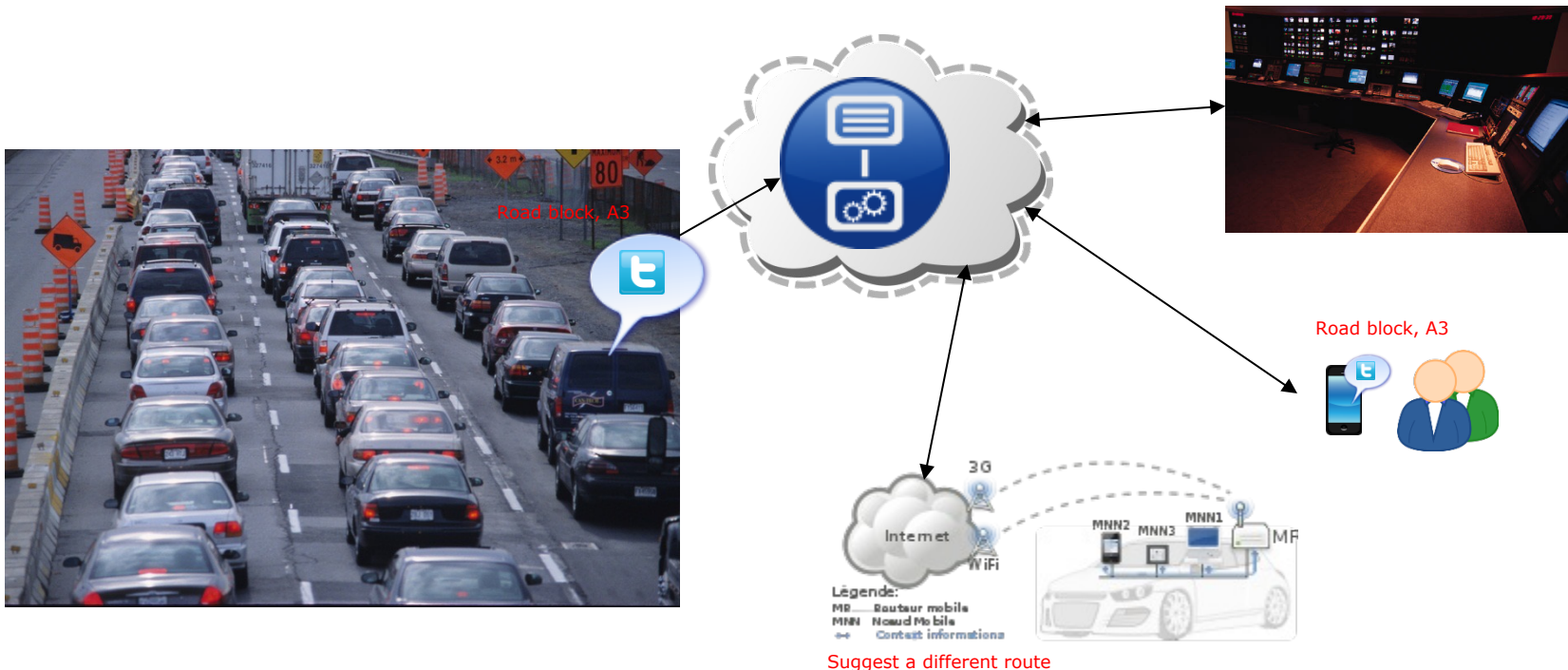
The feedback loop

- A feedback loop is a common and powerful tool when designing a control system.
- Feedback loops take the system output into consideration, which enables the system to adjust its performance to meet a desired output response.



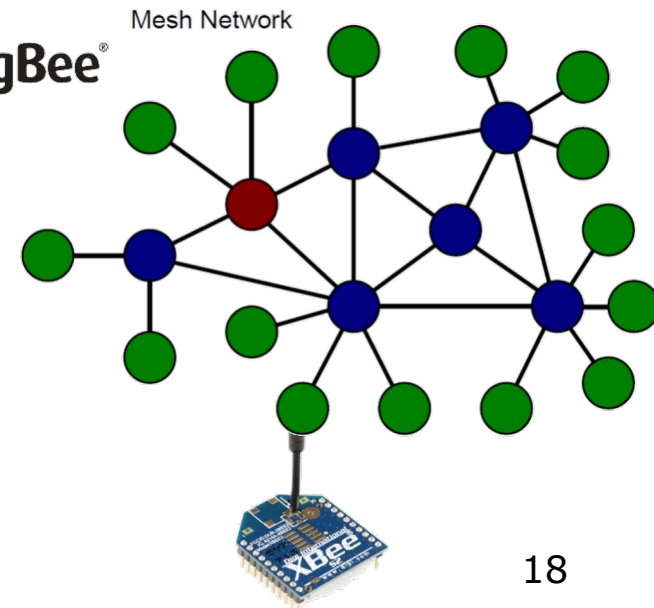
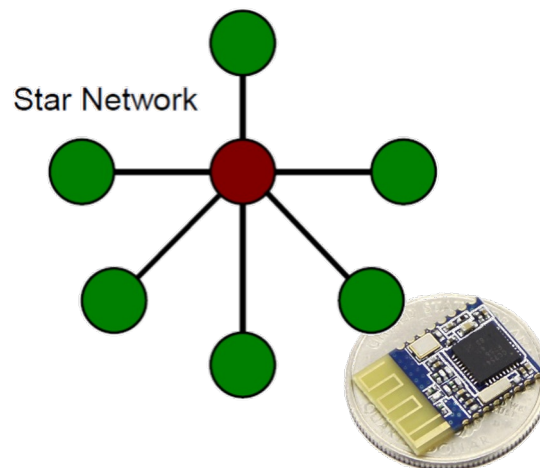
Beyond conventional sensors

- Human as a sensor (citizen sensors)
 - e.g. tweeting real world data and/or events
- Software sensors
 - e.g. Software agents/services generating/representing data



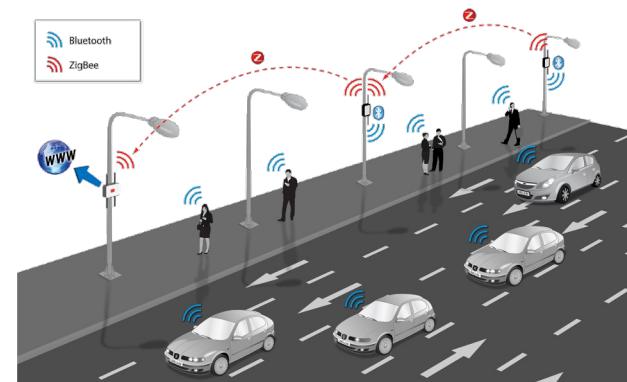
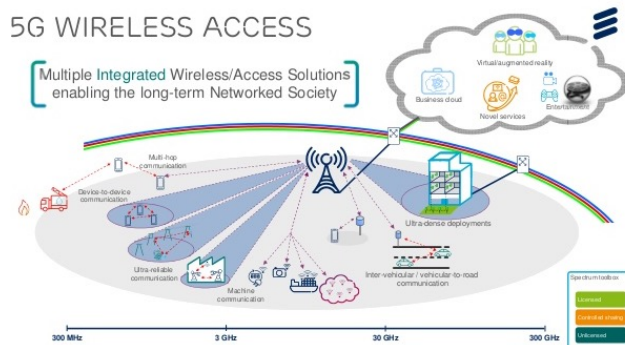
Communications

- The Role of Communications
 - Providing a data link between two nodes
- Communication type:
 - Wireline (e.g. copper wires, optical fibers)
 - Wireless (e.g. RF-based is the most popular choice **(and our focus)**)
- Popular RF-based communication solutions for WLAN or PAN:
 - IEEE 802.15.4
 - IEEE 802.11 (or Wifi)
 - Bluetooth
 - NFC (e.g. RFID)



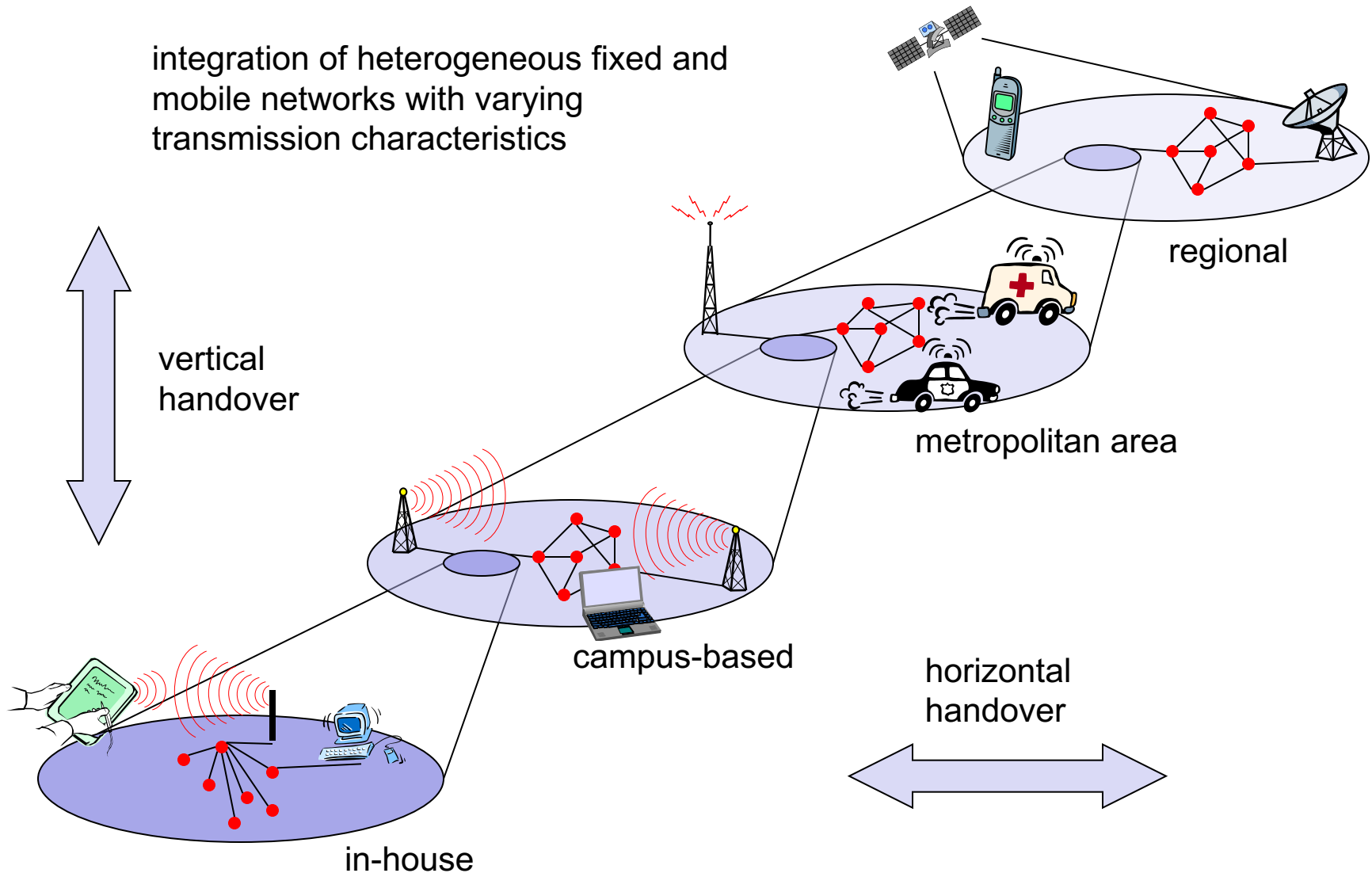
Networks

- The Roles of Networks
 - Managing nodes (discovery, join, leave, etc).
 - Relaying data packets from the source to the destination node
- Networks are a distributed system. All nodes need to perform networking related tasks.
- RF-based Network in IoT is usually Wireless Multi-hop:
 - Wireless Sensor Networks (WSNs)
 - Mobile Wireless Ad hoc Networks (MANETs)
 - Wireless Mesh Networks (WMNs)
 - Vehicular Ad Hoc Networks (VANETs)
- Main concern: Reliability & Performance



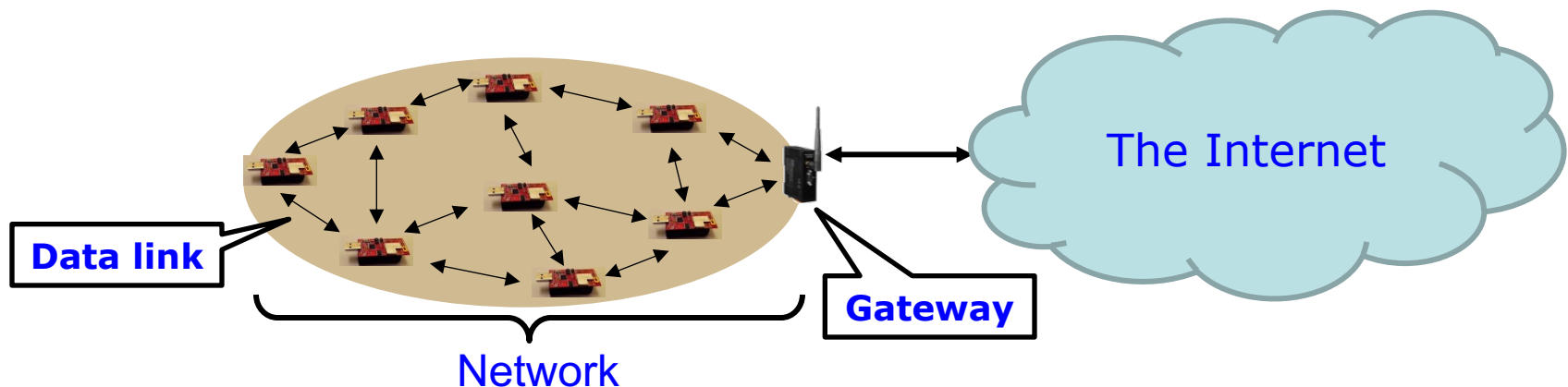
Overlay Networks - the global goal

integration of heterogeneous fixed and mobile networks with varying transmission characteristics



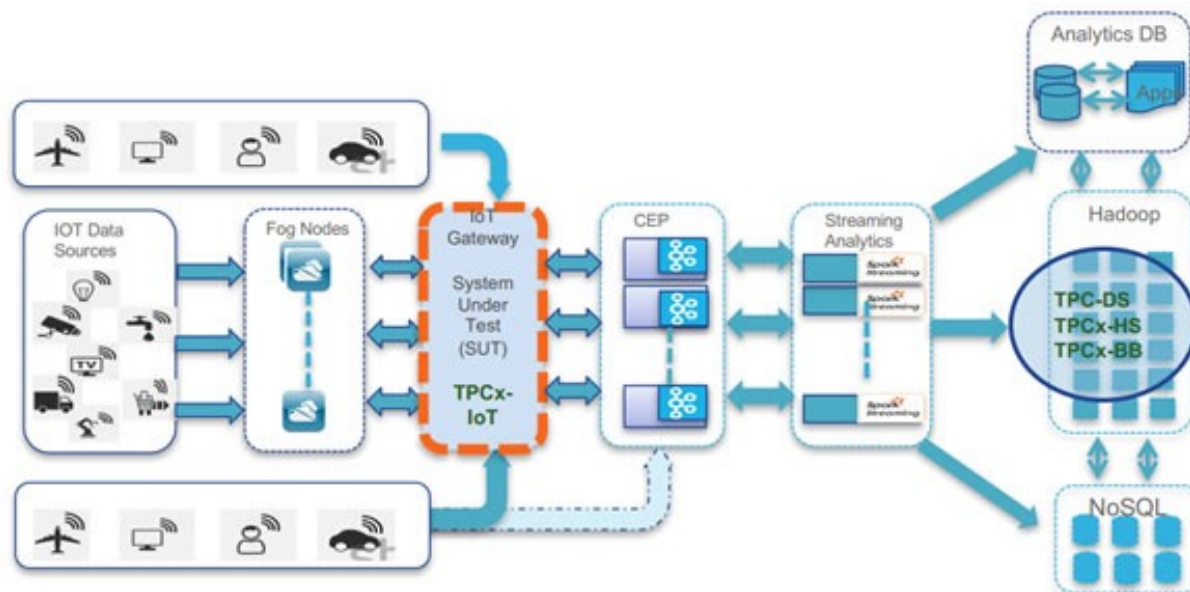
Internet Gateways

- IoT objects connect to resources available over clouds through the Internet
- The Internet uses TCP/IP. This implies that things must also support TCP/IP.
- Gateways (or sinks)
 - For a practical deployment, a gateway is often needed in a network.
 - It offers relaying packets between the network and the Internet.



Gateways and middleware

- It is unlikely that everything will be IP enabled and/or will run IP protocol stack
- Gateway and middleware solutions can interfaces between low-level sensor island protocols and IP-based networks.
- The gateway can also provide other components such as QoS support, caching, mechanisms to address heterogeneity and interoperability issues.



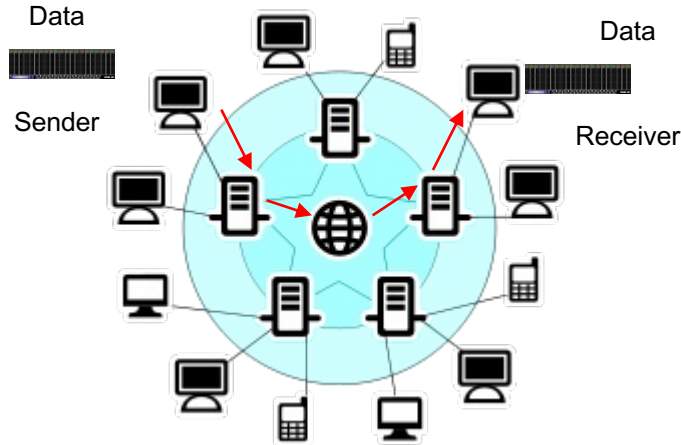
The IoT Protocol Suite

HTTP, Websockets, DNS, XMPP, MQTT, CoAp	Application layer
TLS, SSL	Application Layer (Encryption)
TCP, UDP	Transport Layer
IP(V4, V6), 6LowPAN	Internet Layer
Ethernet, 802.11 WiFi, 802.15.4 (e.g. Zigbee, Thread)	Link Layer

Principles: Layering, modularity, separation of concerns
 Each layer focuses on a particular set of concerns
 and abstracts these concerns from the layer above.

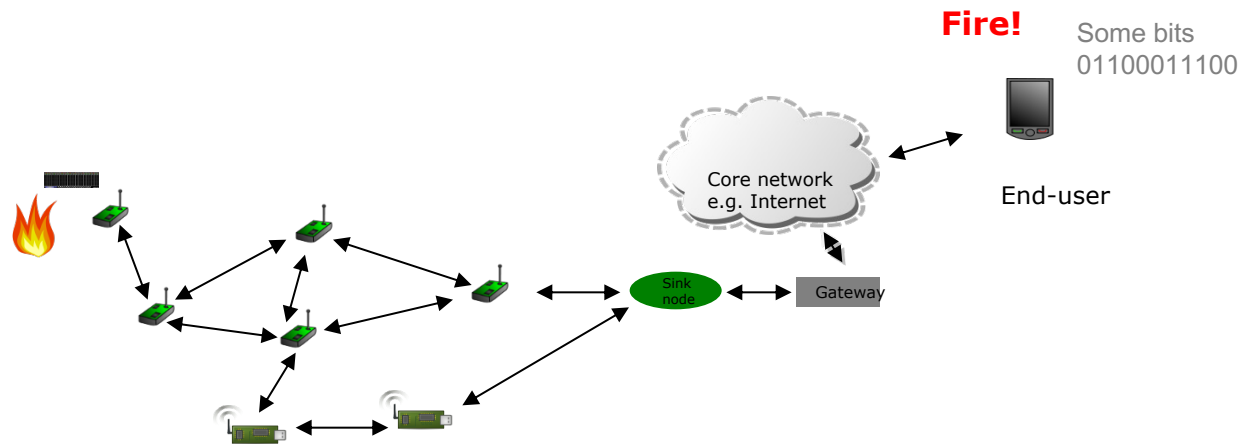
In this course, we will mainly study the application layer.

Example: Type of Services in IoT

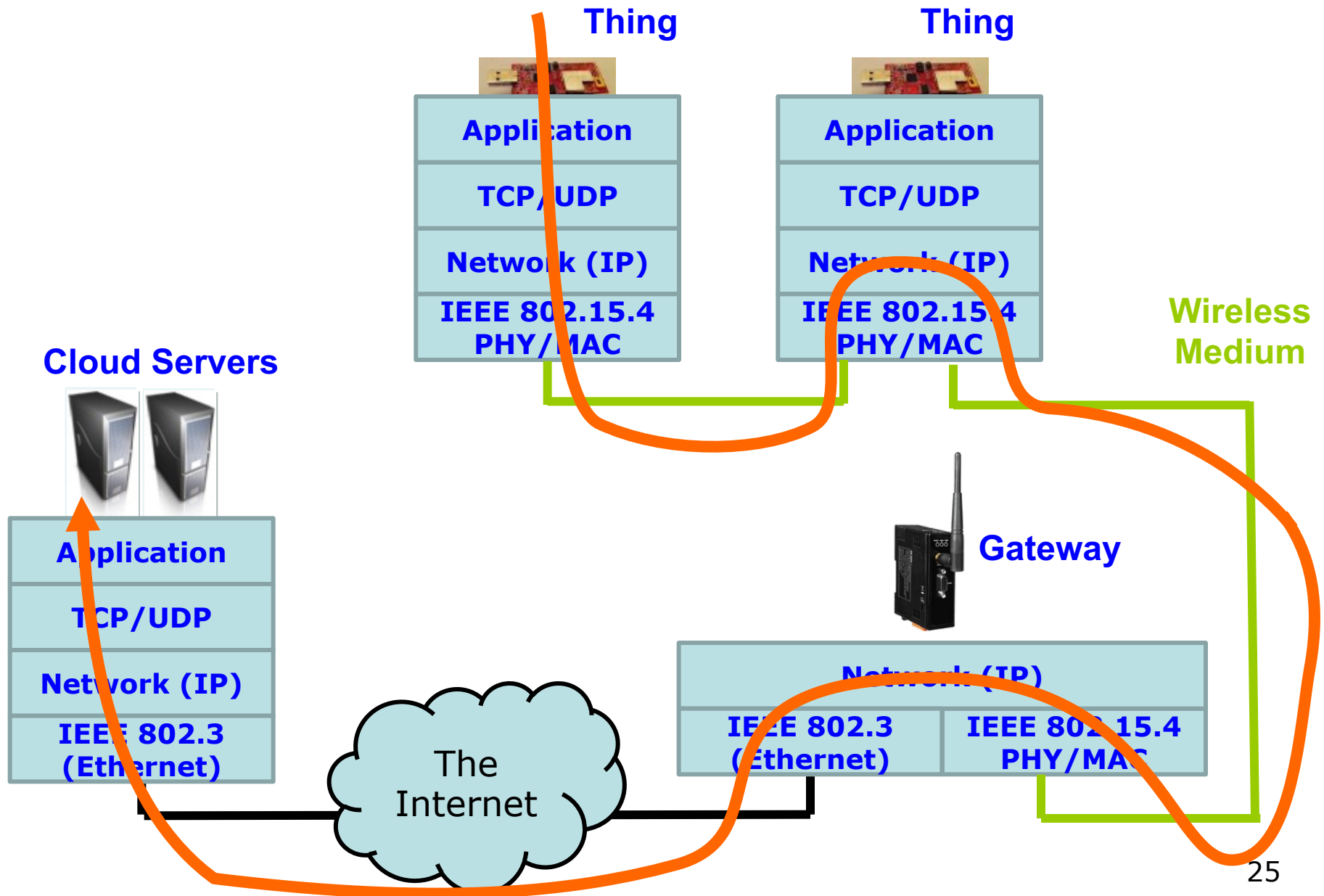


A sample data communication in conventional networks

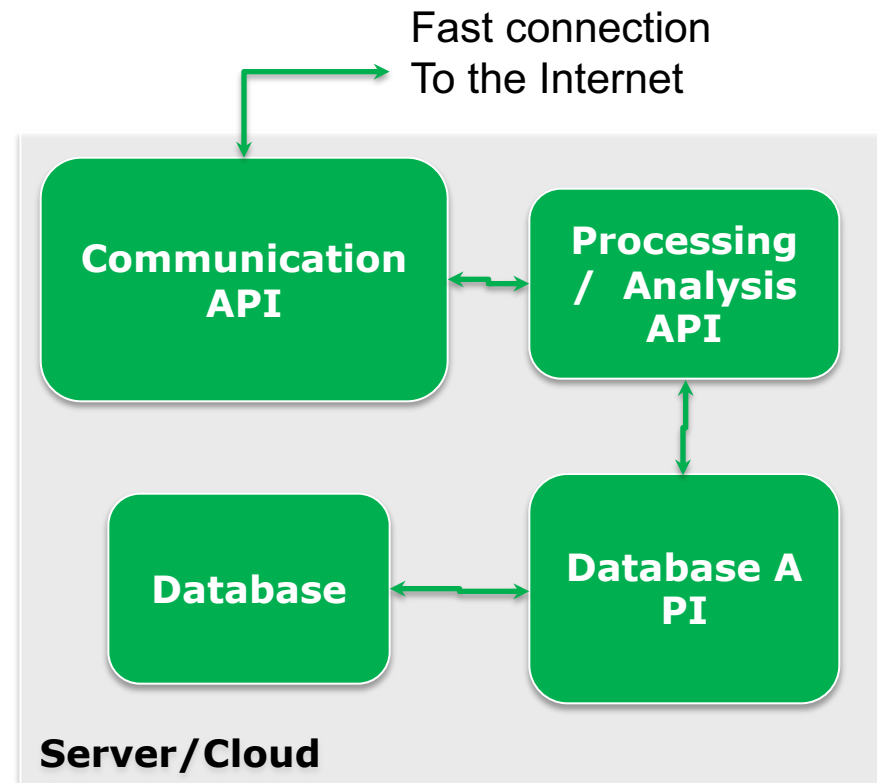
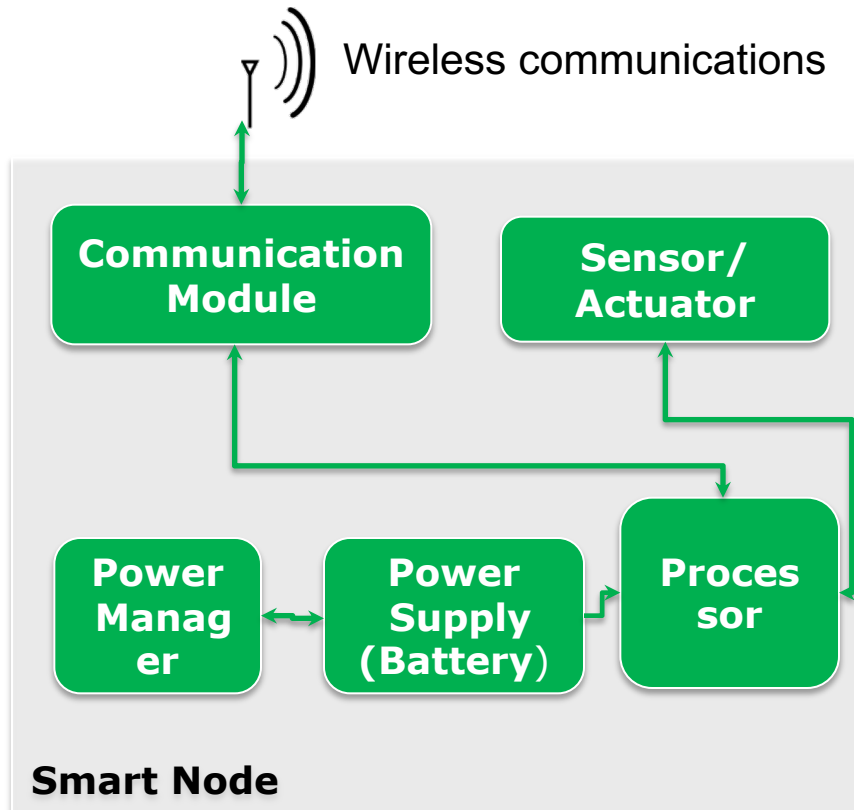
A sample data communication in IoT



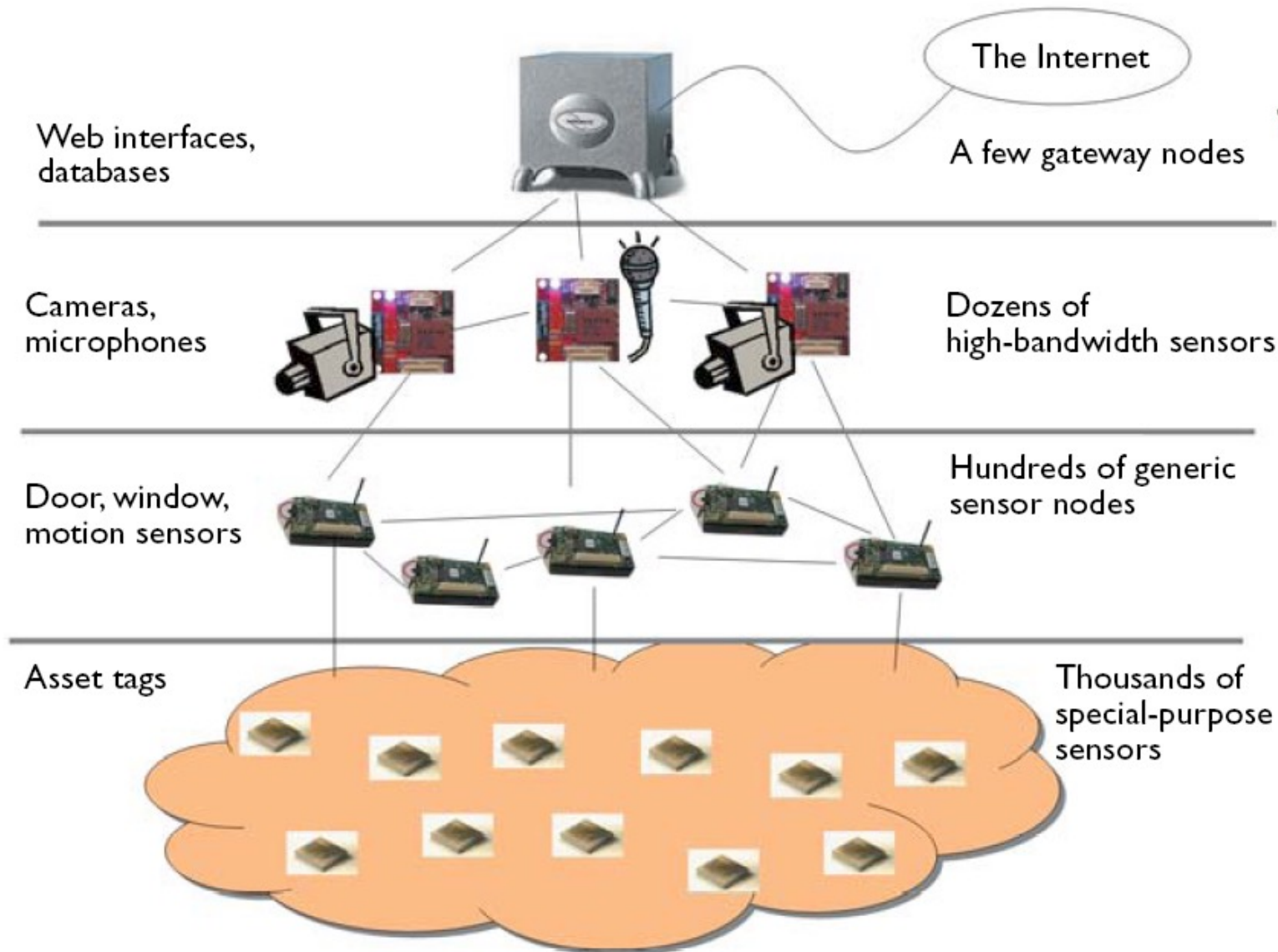
Protocol Stack



IOT Basic Elements Architecture



The IoT ecosystem



Stargate

- 802.11a/b
- Ethernet
- Mica2
- PCMCIA
- Compact-flash
- USB
- JTAG
- RS232